Kian Maleki — Research Scientist | Machine Learning | Scientific Computing | Physics PhD

+1 (737) 600 9803 | kian@drkianmaleki.com | drkianmaleki.com | linkedin.com/in/kian-maleki-phd | github.com/drkianmaleki

University of Iowa (2026) PhD in Physics (GPA: 4.03) | *Creighton University* (2017) MS in Physics (GPA: 4.00)

RESEARCH PROFILE

- Research scientist with a PhD in physics and experience in mathematical modeling, scientific computing, machine learning methodology, and high-dimensional data analysis.
- Developed reproducible computational frameworks for complex physical systems, classifier evaluation, uncertainty/ambiguity analysis, and document intelligence.
- Published peer-reviewed research in **Physical Review B** and **Nano Letters**; current machine-learning research focuses on diagnostic evaluation of binary classifiers beyond global performance metrics.

RESEARCH EXPERIENCE

University of Iowa | Graduate Research Assistant, Physics **Aug 2018 – Aug 2025**

- Developed high-dimensional numerical models for complex physical systems, combining statistical inference, optimization, and large-scale simulation.
- Optimized computational workflows, reducing simulation runtime from days to under one hour through algorithmic improvements and efficient implementation.
- Analyzed noisy, high-dimensional experimental and simulation data to identify structure, extract signal, and validate theoretical predictions.
- Contributed to peer-reviewed publications in condensed matter physics, quantum materials, and solid-state integration of quantum devices.

Creighton University | Graduate Research Assistant, Physics **Aug 2015 – Aug 2018**

- Built simulation-based models for physical systems using Python and Fortran, including N-body simulations and molecular dynamics.
- Designed parallelized computation pipelines for large-scale simulations, improving scalability with HPC resources.
- Developed interactive scientific software tools for simulation and analysis, enabling user-driven exploration of model behavior.

Creighton University | Assistant Professor of Physics **Aug 2025 – Present**

- Teach undergraduate physics courses with emphasis on quantitative reasoning, modeling, optics, electronics, and applied problem-solving.
- Developed interdisciplinary lab curriculum integrating AI tools and chatbots into scientific workflows while emphasizing responsible and ethical use.
- Supervise and mentor teaching assistants, improving instructional consistency and student support across large-enrollment courses.

MACHINE LEARNING AND APPLIED AI RESEARCH

Ambiguity Range Framework **2026**

Tech: Python, Scikit-Learn, XGBoost, PyTest, Statistical Testing, Model Evaluation

link: https://github.com/drkianmaleki/ambiguity-framework

- Developed a model-agnostic diagnostic framework for evaluating binary classifiers beyond global metrics such as AUC-ROC.
- Introduced ambiguity-focused metrics that quantify prediction concentration near decision boundaries and threshold-sensitive degradation in precision, recall, and F1.
- Built reproducible experiments across real-world datasets with calibration analysis, bootstrap confidence intervals, paired statistical tests, and automated test coverage.

PDF Document Intelligence Extractor **2026**

Tech: Python, NLP, Regex, Pandas, PDF Text Extraction, RAG

link: https://github.com/drkianmaleki/PDF-Document-Intelligence-Extractor

- Built a document intelligence pipeline to extract structured regulatory information from PDFs using native text-layer parsing.
- Developed rule-based and NLP extraction methods to identify, normalize, and structure information from semi-structured documents.
- Designed the project as part of a broader document intelligence research direction that includes advanced OCR workflows for scanned or image-heavy documents.

RAG-Powered Knowledge Assistant **2026**

Tech: Python, PyTorch, Transformers, NLP, Docker

link: https://github.com/drkianmaleki/RAG-Powered-Knowledge-Assistant

- Built a local retrieval-augmented generation system for document question answering with context-aware responses.
- Designed chunking and retrieval strategies to preserve semantic continuity across long documents and reduce unsupported responses.
- Containerized the application with Docker for reproducible deployment and local execution.

TECHNICAL EXPERTISE

- **Research Methods:** Mathematical modeling, statistical inference, uncertainty analysis, hypothesis testing, bootstrap confidence intervals, paired statistical tests, Monte Carlo simulation.
- **Machine Learning:** Scikit-Learn, XGBoost, PyTorch, TensorFlow, Transformers, calibration analysis, model evaluation, NLP, RAG, MLflow.
- **Scientific Computing:** Python, NumPy, Pandas, Polars, Numba, Fortran, Linux/Bash, HPC, MPI, multithreading, numerical optimization.
- **Engineering and Reproducibility:** Git, GitHub Actions, pytest, Docker, DVC, FastAPI, Pydantic, AWS S3/EC2, reproducible experiment design.

AI AND MACHINE LEARNING TRAINING

TripleTen | AI & Machine Learning Program **2026**

- Completed project-based training in supervised and unsupervised learning, deep learning, NLP, computer vision, feature engineering, and model evaluation.
- Built end-to-end machine-learning projects using Scikit-Learn, Python, SQL, Git, and modern ML workflows.

Extern, Pfizer | AI-Powered Document Intelligence Externship **2026**

- Developed AI-powered document intelligence workflows for healthcare documents using native PDF text extraction, advanced OCR workflows, LLMs, and RAG architectures.
- Built retrieval-augmented pipelines to improve accuracy and reduce hallucinations in document-based question answering systems.

SELECTED PUBLICATIONS

- *Maleki, K. et al.* “Crystal fields, exchange and dipolar interactions...” **Physical Review B**, 2025.
- *Maleki, K. et al.* “A General and Modular Approach to Solid-State Integration...” **Nano Letters**, 2025. (Related to a patented quantum device)
- *Maleki, K.* “The Ambiguity Range Framework: A Diagnostic Toolkit for Operational Evaluation of Binary Classifiers.” Manuscript under review, 2026.